

Final Project

Predicting Alcoholism Using Various Characteristics

Nikita Park, Jeremy Shiu, Ian Turner, Mingchen Wang, Josh Xu

University of California, Los Angeles

STATS 101C

Professor Akram Almohalwas

December 17th, 2023

Abstract

The goal of this project was to predict whether or not the subjects in the provided testing data were alcoholics, based on certain vital and lifestyle characteristics, by training a model on a more complete data set. This paper discusses the journey that brought us to the final model and evaluates the final model itself. Our final model used the XGBoost algorithm and utilized all the predictor variables in the data set. Our accuracy was 73.38%, good for 4th place in lecture 2 and 5th place overall.

1. Introduction

Alcohol is consumed in places across the world in various forms and is very important economically and culturally to many countries. In the United States alone, sales of alcohol accounted for nearly \$90 billion in gross revenue in 2023 [1]. However, alcohol is well known to have negative effects, such as inhibiting mental functions and causing heart problems, liver damage, and even cancer [2], causing approximately 141,000 deaths in the United States every year [3]. Therefore, it is important to understand which factors are associated with alcohol abuse to determine who is most at risk.

In this paper, we recap our journey of performing exploratory data analysis, creating increasingly complex models, and fine-tuning parameters to arrive at our best model. We also reflect on the limitations of our methodology and discuss our findings.

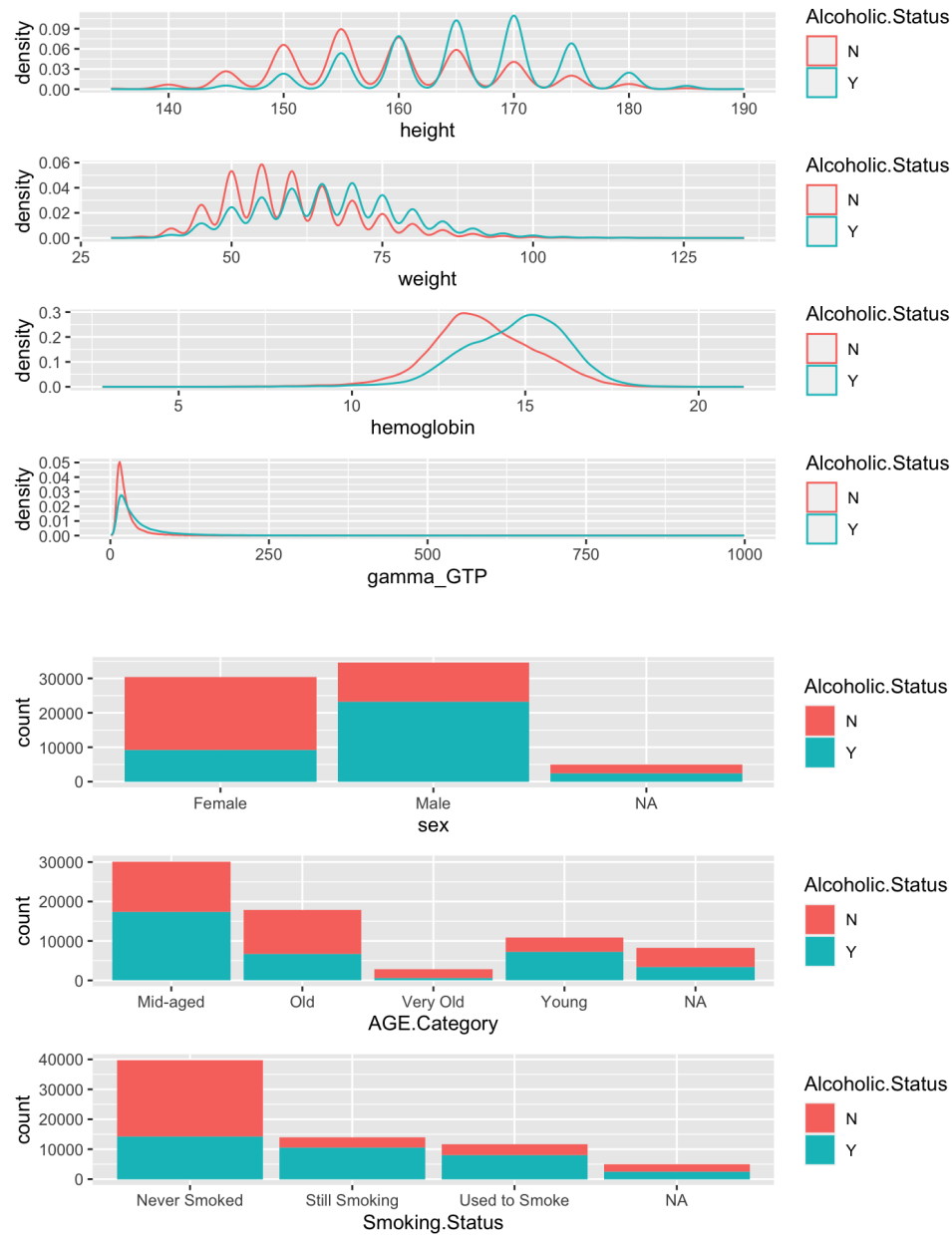
2. Exploratory Data Analysis

A training data set and a testing data set were provided for this project. The training data set contained 70,000 observations with 27 variables, including the alcoholic status response variable. The testing data had 30,000 observations with 26 variables, omitting the alcoholic status of the patients. The variables include vital characteristics like age, sex, and weight as well as lifestyle habits like smoking status.

Upon receiving the data, our first course of action was to perform exploratory data analysis (EDA) on the training data. The training data consisted of 34,887 alcoholics (49.84%) and 35,113 non-alcoholics (50.16%) for a roughly 50–50 split.

We created density plots for each numerical predictor and side-by-side bar plots for each categorical predictor to see if they had any clear association with alcoholism. Through EDA, we were able to see that hemoglobin and smoking status were particularly strong predictors, but no single predictor was sufficiently strong on its own. A basic logistic regression model using only hemoglobin as a predictor had a testing accuracy of only 63.93%.

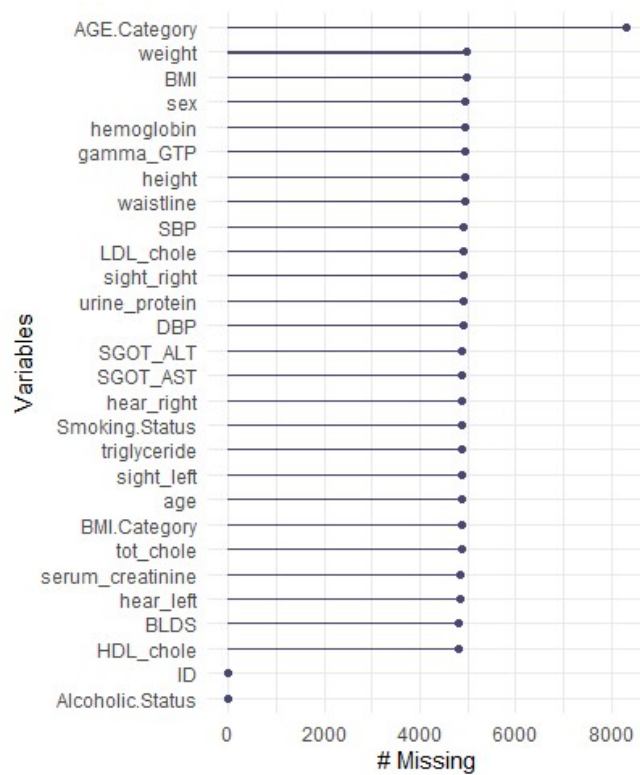
Figure 1: Plots of the Strongest Predictors in Relation to Alcoholic Status



The biggest issue we found during our EDA was an enormous amount of missing values. The missing values were randomly distributed such that we could not be comfortable deleting any predictors or any observations. This meant we had to impute missing values, a

computationally tedious procedure that took us several days to try several multiple imputation algorithms from four R packages—Amelia, Hmisc, mice, and missForest. We found that the highest testing accuracies came from models trained from missing values imputed by Hmisc’s aregImpute function.

Figure 2: Distribution of Missing Values by Predictor



The aregImpute function in R utilizes bootstrapping, resampling an original data set to create multiple simulated samples [4], to “approximate the process of drawing predicted values from a full Bayesian predictive distribution” [5]. While the aregImpute function in R has several parameters, we essentially chose to use the default settings with `n.impute = 5` and the method of

predictive means matching since the other available method, regression, would not work with the categorical variables in the data set. Using a higher `n.impute` value, which controls the number of times the process is run, could theoretically lead to more accurate imputations, but issues with computation times and hardware limitations led us to pick a smaller value.

3. Methods and Models:

Initial Models:

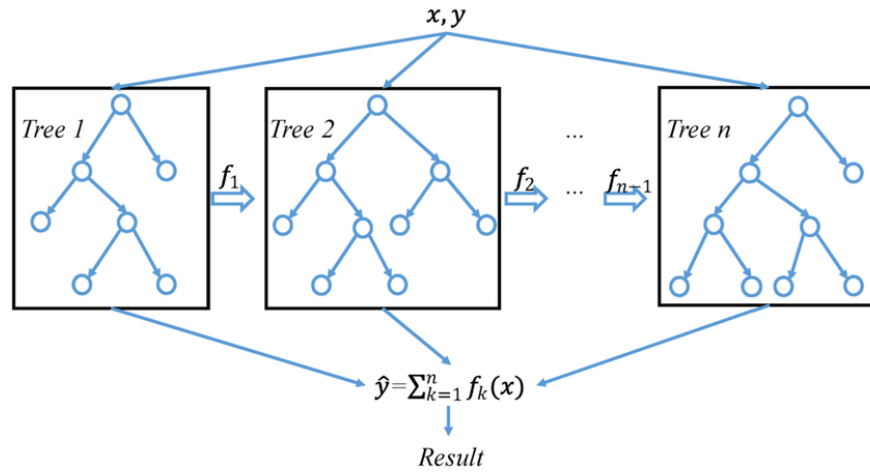
Since we were originally limited to ten Kaggle submissions per day, we did not want to waste submissions with models that had very low prospects of success. We did this by validating our models on the training data set before running them on the testing data—either by running our models on the entire training set, performing an 80–20 data split within the training set, or by performing 10-fold cross-validation and bootstrapping. Through this process, we found that some methods like linear discriminant analysis (LDA) and quadratic discriminant analysis (QDA) had comparatively terrible results even when run on the training data, so we did not consider them for our final model.

Best Algorithm: XGBoost

Realizing our model is not a linear fit from LDA, we decided that a tree model would be better at predicting the accuracy. We chose to use XGBoost because of its widespread recognition as the most accurate tree algorithm [6]. Indeed, our initial XGBoost algorithm with no variable selection, no cross-validation, and no pruning produced an accuracy of 72.283%, far

superior to any of our previous models. Thus we quickly settled on XGBoost as our algorithm of choice, electing to spend most of 2 weeks fine-tuning its parameters to create the best model.

Figure 4: XGBoost Diagram



[7]

Although having been developed quite recently, XGBoost builds off of earlier decision tree algorithms, making it straightforward to interpret. Similarly to random forest, XGBoost builds numerous decision trees based on the instance, as seen in Figure 3. Rather than building these trees independently and randomly, each of the decision trees in XGBoost is learning from the residuals of the previous ones. This is called boosting, where XGBoost gets its name. In XGBoost, a specific boosting algorithm called gradient boosting is applied, where the loss function is minimized using a gradient descent [8]. The loss function calculates the difference between the actual value and the predictions made by existing trees. This loss function can be defined by users, which we chose to be the default method of negative log-likelihood. The sum

of the results from the trees becomes the predictive output in XGBoost, as opposed to using a majority vote as in random forest.

Resampling and Fine Tuning:

From our initial XGBoost model, we improved our accuracy to 73.226% by performing 10-fold cross-validation. We then found that using bootstrapping instead of cross-validation as our resampling method increased accuracy to 73.326%.

The `train()` function in R's `caret` package enabled us to fine-tune our parameters directly. As part of the bootstrapping process, the best values for the parameters are found via multiple training and testing. Different combinations of parameters are tested in each boosting iteration. A graph depicting the change in resampling accuracy with different parameters is given in Figure 4.

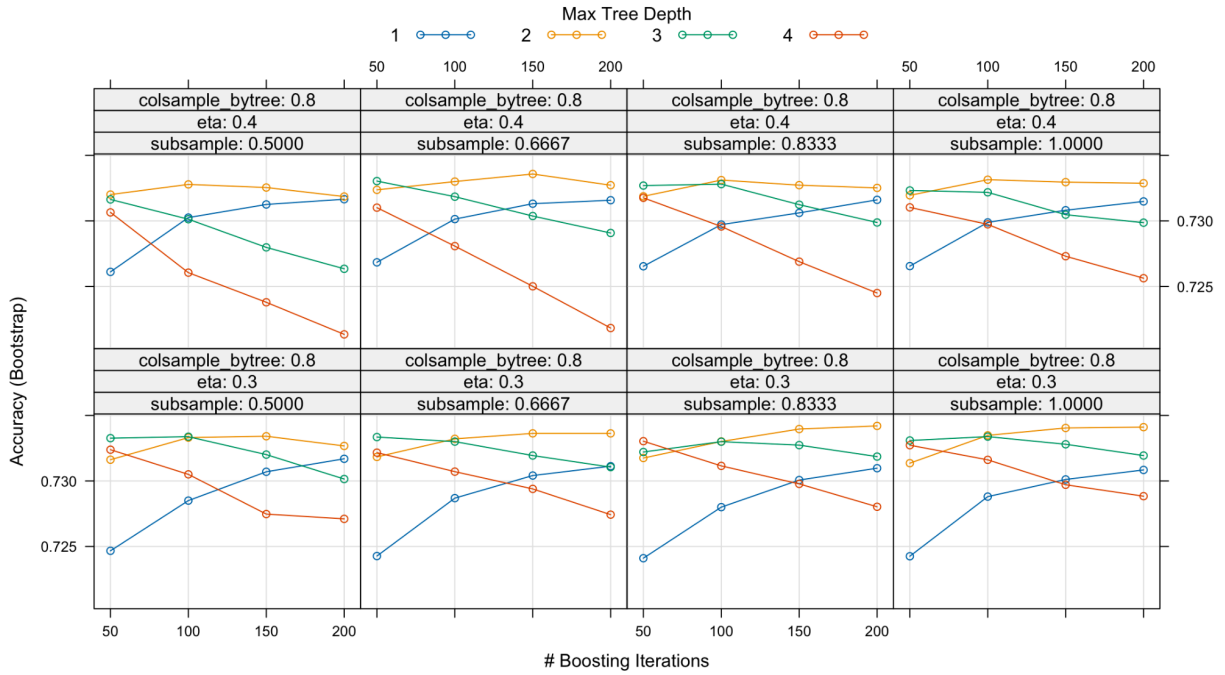
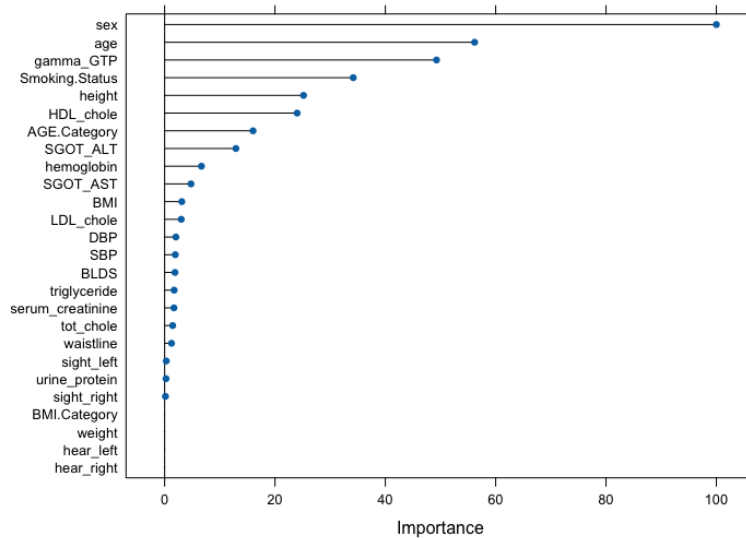
Figure 4: XGBoost Best Fit Plot

Figure 4 shows that pruning the maximum depth of our trees to 2 tends to improve bootstrap accuracy. Interestingly, the learning rate (η), the value that affects the model's learning rate on fitting to the residual errors, had the best accuracy when set at the default rate of 0.3. Generally, a rate of 0.01–0.1 is chosen since smaller values can improve generalization [9]. We attempted to fine-tune the learning rate by hand, but we did not see an improvement in accuracy.

Through the fine-tuning process, we arrived at a model with trees pruned to a maximum depth of 2, with all other parameters unchanged from R's default values. This took us to an accuracy of 73.363%, which was good for 3rd place at the time of submission.

Further Attempts

Figure 5: Variable Importance Plot

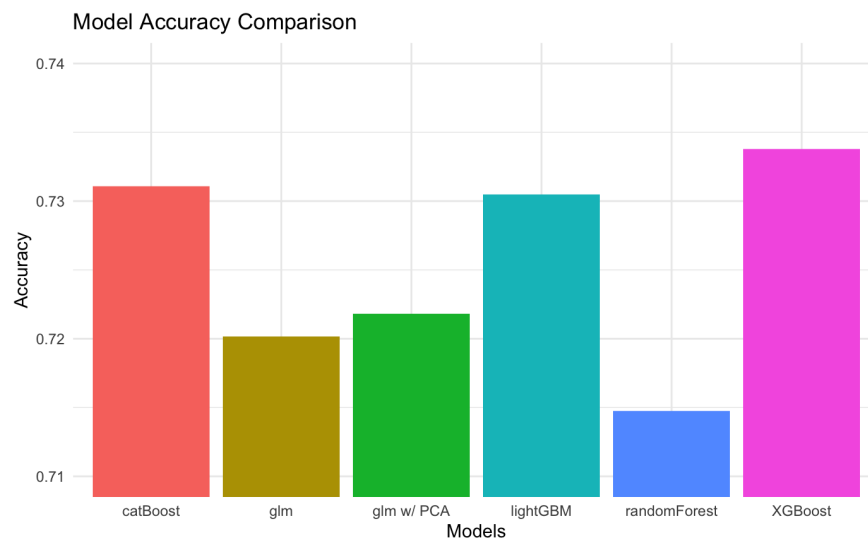


Our next step would naturally have been to perform variable selection to address overfitting. Figure 5 shows the importance of each predictor variable in the XGBoost model. We fitted several models that dropped the weakest twenty predictors, ten predictors, five predictors, two predictors, and one predictor, but our accuracy dropped each time. This was a somewhat surprising result since we understand that the predictor BMI category is completely redundant with the predictor BMI, and we believe that hearing ability has nothing to do with alcoholism whatsoever from an intuitive perspective. Yet even when these useless or seemingly useless predictors were removed, our accuracy went down, so our final model included all the predictors.

This issue will be addressed in the Discussion and Limitations section of this paper, but here we will note that we would have chosen a smaller model removing the obviously redundant predictors if our goal was to produce a model that would be used to make real-world predictions.

However, since the goal of this project was to achieve the highest possible score on Kaggle, we went with the full model.

Figure 6: Model Accuracy Comparison



We also attempted several other algorithms, namely CatBoost, LightGBM, random forest, and logistic regression with principal component analysis (PCA) performed on the predictors, but none of these algorithms produced a better accuracy than XGBoost. As previously mentioned, we did not bother trying LDA or QDA because their accuracies when performed on the training data were too low. We also attempted to use artificial neural networks (ANN) in the final days of the project, but it was too computationally intensive to produce a usable result.

As the end of the project approached, we gradually fell from 3rd place to 6th place. Desperately hoping to climb up the leaderboard and unable to fine-tune our model any further,

we attempted to increase our accuracy through less academically rigorous means like trying different seed values for random number generation. On the final day of the project, we were able to increase our accuracy from 73.363% to 73.38%, good for 5th place in the competition and 4th place within Lecture 2. As we will discuss later in this paper, we would never have done this if our model was to be used in a serious academic or medical setting.

We finished the project with 173 submissions, the most of any group by a whopping 44 submissions.

4. Discussion and Limitations:

While our model was very successful with its 5th place finish, we acknowledge a wide range of limitations, most of which are due to the structural design of the Kaggle competition. Our goal was not to create the best model for predicting alcoholic status given our 26 predictor variables. It was to create the best model for predicting the alcoholic status of 30,000 specific individuals given 26 predictor variables. Accordingly, we engaged in several practices that would have been highly inappropriate in a serious academic or medical setting, such as using all 26 predictors and attempting to increase accuracy by choosing different seeds for random number generation. We understand that such practices are indefensible in a real-life context. If this were an actual study, we would have removed several predictors from the outset—namely BMI Category, Weight, and Age Category for being redundant with other predictors, and the eyesight and hearing predictors for having nothing to do with alcoholism.

One limitation of this entire project, not specific to our model, is that it merely predicts alcoholism based on its association with other factors, without distinguishing whether these

factors *cause* or *are caused by* alcoholism. For instance, there is evidence that higher cholesterol levels and cardiovascular issues *are caused by* alcoholism [9], but if the objective of this model were to identify individuals *at risk of* developing alcoholism based on their vital characteristics, then the cholesterol variables should not be used to *predict* alcoholism. In practical terms, this means our results should not be used to promote the consumption of less fatty foods; while eating less fatty foods will surely decrease cholesterol, it will most likely not decrease one's risk of alcoholism. If this were a real study, we would have considered the objectives and performed variable selection accordingly. Various other predictors may also have confounding variables. For example, smoking and alcoholism are well known to be correlated [10], but it is unclear whether one or both causes the other or if they are both caused by other confounding variables.

Other variables may need to be removed depending on the context. For instance, there is a noticeably lower rate of alcoholism among older people than younger people, but this might be attributable to survivorship bias since alcoholics are less likely to survive to old age. Another example is Gamma GTP, which is a liver enzyme that suggests liver damage when present in high concentrations in the bloodstream [11]. This predictor would be useful for identifying current alcoholics, but not useful for predicting whether somebody is at risk of developing alcoholism.

Our model also produced some interesting results that may not be intuitive. In particular, sex was the most important variable in our final model (Figure 5). Our data show that men have much higher rates of alcoholism than women (Figure 1), which we speculate might result from social pressures. This seems to warrant further investigation. The literature shows that this

phenomenon is not unique to our data set but is actually a rather common global phenomenon [12], despite men generally having a higher tolerance for alcohol than women [13].

As for the technical side of this project, our most notable limitation was that certain algorithms took an extremely long time to run. There were a few instances where some algorithms, especially some of the imputation methods, may have been more effective with more iterations. This was simply infeasible given our computer speed and time constraints. This provides some insight into both the necessity for more powerful computers and, more relevantly for this project, the importance of reducing the complexity of models and model calculation methods where possible. A major strength of our model is that it uses only two trees, making it significantly computationally easier and quicker to run as well as easier to interpret, all while performing better than more complex models with four or five trees. This demonstrates that difficulties with computational power can often be overcome with ingenuity or can be avoided entirely because simpler models can be just as good, if not better.

5. Conclusion:

Because there is no "right" model for every situation and every dataset, the process of finding the best model is not a simple one; it requires experimentation, trial and error, and persistence to unlock the secrets in the data. This competition showed that even with thousands of observations, two completely different models and methods can get within thousandths of a percentage point of each other. Model selection is more than finding the model with the highest accuracy, but finding one that efficiently and effectively makes predictions. Throughout the process of exploring the various models, imputation techniques, and variable selection for this

Kaggle competition, we learned valuable skills that can be applied to real-world situations. With multiple stages in the project, from variable exploration to investigating potential correlations, to filling in missing values intelligently, to fitting the most accurate model, we have been able to apply the many methods learned throughout this course. If given a large dataset, we can explore it with similar methods to create predictive tools that can assist people with diagnosing health concerns, predicting an individual's likelihood of some condition given other factors in their life, and more.

References

1. Jan Conway, and Feb 16. "U.S. Revenue of Alcoholic Beverages 2022." *Statista*, 16 Feb. 2023, www.statista.com/statistics/237868/us-revenue-of-alcoholic-beverages-by-type/.
2. "Alcohol's Effects on the Body." *National Institute on Alcohol Abuse and Alcoholism*, U.S. Department of Health and Human Services, 21 Oct. 2021, www.niaaa.nih.gov/alcohols-effects-health/alcohols-effects-body.
3. "Alcohol Abuse Statistics [2023]: National + State Data." *NCDAS*, National Center for Drug Abuse Statistics, 1 Jan. 2023, drugabusestatistics.org/alcohol-abuse-statistics/.
4. Joseph, Trist'n. "What Is Bootstrapping Statistics?" *Built In*, 11 Sept. 2023, builtin.com/data-science/bootstrapping-statistics.
5. Harrell, Frank. "Aregimpute: Multiple Imputation Using Additive Regression, Bootstrapping, and Predictive Mean Matching." *RDocumentation*, www.rdocumentation.org/packages/Hmisc/versions/5.1-1/topics/aregImpute. Accessed 16 Dec. 2023.
6. "What Is Xgboost? An Introduction to XGBoost Algorithm in Machine Learning: Simplilearn." *Simplilearn.Com*, Simplilearn, 7 Nov. 2023, www.simplilearn.com/what-is-xgboost-algorithm-in-machine-learning-article.
7. Mingtao, Li. "A General Architecture of XGBoost | Download Scientific Diagram." *ResearchGate*, Aug. 2019, www.researchgate.net/figure/A-general-architecture-of-XGBoost_fig3_335483097.

8. Brownlee, Jason. "A Gentle Introduction to the Gradient Boosting Algorithm for Machine Learning." *MachineLearningMastery.Com*, 14 Aug. 2020, machinelearningmastery.com/gentle-introduction-gradient-boosting-algorithm-machine-learning/.
9. Sen, Vishwajit. "Gradient Boosting and XGBoost Explained in Detail." *LinkedIn*, 29 June 2023, www.linkedin.com/pulse/gradient-boosting-xgboost-explained-detail-vishwajit-sen/.
10. Shiffman, Saul, and Mark Balabanis. "Do Drinking and Smoking Go Together?" *Alcohol Health and Research World*, U.S. National Library of Medicine, 1996, www.ncbi.nlm.nih.gov/pmc/articles/PMC6876501/.
11. "Gamma-Glutamyl Transferase (GGT) Test: What It Is & Results." *Cleveland Clinic*, 9 Nov. 2021, my.clevelandclinic.org/health/diagnostics/22055-gamma-glutamyl-transferase-ggt-test.
12. White, Aaron M. "Gender Differences in the Epidemiology of Alcohol Use and Related Harms in the United States." *National Institute on Alcohol Abuse and Alcoholism*, U.S. Department of Health and Human Services, 29 Oct. 2020, arcr.niaaa.nih.gov/volume/40/2/gender-differences-epidemiology-alcohol-use-and-related-harms-united-states.
13. "Excessive Alcohol Use and Risks to Women's Health." Centers for Disease Control and Prevention, October 17, 2022. <https://www.cdc.gov/alcohol/fact-sheets/womens-health.htm>.